

Kali Attacker Routing Configuration

Purpose

The Kali attacker VM simulates external adversary traffic entering the Virtual SOC environment. For realism and full visibility, **all Kali → victim traffic must pass through pfSense**, ensuring: - Suricata NIDS can inspect packets - pfSense firewall can enforce policy - Wazuh SIEM receives complete correlated logs

Because the internal-net (105.45.5.0/24) is isolated behind pfSense, Kali must be configured with **persistent static routes** directing all RFC1918 and SOC subnets through the transit-net gateway.

1. Attacker VM Placement

The Kali instance is attached to **ext_net**, appearing as an external threat actor.

Example ext_net IP:

172.17.184.110

Default gateway is OpenStack's floating IP infrastructure. Kali cannot directly reach 105.45.5.0/24 without explicit routing rules.

2. Routing Requirements

The following networks must route through **pfSense WAN (10.250.2.5)**: - SOC transit-net: 10.250.2.0/29
- Internal network: 105.45.5.0/24 - (Optional) Additional isolated/lab networks: 10.10.15.0/24

These routes ensure: - All packets traverse pfSense - Suricata sees scans, exploits, lateral movement - Internal hosts can return traffic correctly

3. Kali Routing Configuration

Kali uses the **legacy ifupdown** networking model. Persistent routes must be added to `/etc/network/interfaces`.

Edit file:

```
sudo vim /etc/network/interfaces
```

Configured version:

```
#START OF FILE
source-directory /run/network/interfaces.d

auto eth0
iface eth0 inet dhcp
    # persistent routes:
    post-up ip route replace 10.250.2.0/29 via 172.17.184.110 dev eth0
    post-up ip route replace 105.45.5.0/24 via 172.17.184.110 dev eth0
    post-up ip route replace 10.10.15.0/24 via 172.17.184.110 dev eth0

    # clean up on down:
    pre-down ip route del 10.250.2.0/29 via 172.17.184.110 dev eth0 || true
    pre-down ip route del 105.45.5.0/24 via 172.17.184.110 dev eth0 || true
    pre-down ip route del 10.10.15.0/24 via 172.17.184.110 dev eth0 || true
#END OF FILE
```

Apply changes:

```
sudo ifdown eth0 && sudo ifup eth0
```

4. Routing Logic Explanation

Why send traffic to 172.17.184.110?

This is the **OpenStack ext_net gateway** for the Kali VM. From there, traffic follows this chain:

```
kali → ext_net → ex-router → transit-net → pfSense WAN → internal-net
```

Why not route directly to pfSense?

Because pfSense WAN (10.250.2.5) is **not directly reachable** from Kali—it sits behind the ext_net router. Therefore, the ext_net gateway must be used as the next hop.

5. Validation

Test 1 — Ping pfSense WAN

```
ping 10.250.2.5
```

Test 2 — Ping Internal Hosts

```
ping 105.45.5.1    # pfSense LAN  
ping 105.45.5.72  # victim host
```

Both should succeed.

Test 3 — Nmap Scan

```
nmap -sS -T4 -p- 105.45.5.72
```

Suricata should detect: - ET SCAN events - Potential brute-force or auth activity

Test 4 — Exploitation

Use Metasploit or manual payloads. Suricata + Wazuh correlation should show complete attack chain.

6. Troubleshooting

Symptom: No route to host

Check:

```
ip route show
```

Ensure route entries exist.

Symptom: Suricata not detecting scans

Cause: Traffic not routed through pfSense. Fix: Verify the next-hop gateway and ensure ex-router static route exists.

Symptom: Internal hosts unreachable

Fix: Ensure pfSense has NAT enabled on WAN.

7. Summary of Responsibilities

The Kali attacker configuration ensures: - All attack traffic flows through pfSense - Suricata can inspect packets and generate alerts - Wazuh receives correlated logs - Internal hosts remain isolated

Correct routing is essential for a realistic attacker framework within the Virtual SOC environment.